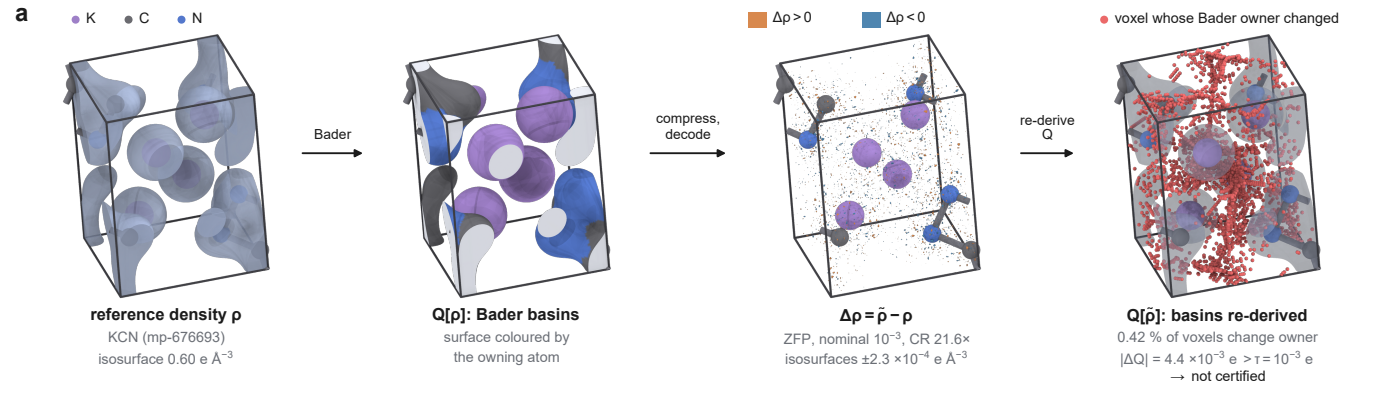

a
K
C
N
Bader
compress, decode
Δρ > 0
Δρ < 0
re-derive Q
voxel whose Bader owner changed
reference density ρ
KCN (mp-676693)
isosurface 0.60 e Å⁻³
Q[ρ]: Bader basins
surface coloured by the owning atom
Δρ = ρ̃ − ρ
ZFP, nominal 10⁻³, CR 21.6×
isosurfaces ±2.3 ×10⁻⁴ e Å⁻³
Q[ρ̃]: basins re-derived
0.42 % of voxels change owner
|ΔQ| = 4.4 ×10⁻³ e > τ = 10⁻³ e
→ not certified
b
Contract
Qualification
Allocation and prediction
Certification
Diagnosis
ρ, Q, τ
QoI, algorithm, tolerance, exact or approximate inputs
ρ + δ_k
probes on the approximate inputs
f_C = max_k |ΔQ_k|
stability floor
f_C < τ
f_C ≥ τ
Δ_G ∝ w(G)^−1/2
step law from the operator symbol, w = |a(G)|²
G_pred
gain over blind allocation, committed before any coding
ρ̃
decoded field (any codec)
ΔQ_i
each analysis re-run on it
|ΔQ_i| < τ
|ΔQ_i| ≥ τ
certified
valid for these QoIs and τ
certificate stored with the field
not certified
error spectrum × operator
→ tighter setting or other codec
non-evaluable
e.g. the Bader partition field
→ change the contract or τ